

Max. Marks: 100

Class: SE

Name of the Course: Computational Mathematics

Course Code: [NON-CBGS]

Duration: 3 Hrs

Semester: IV

Branch: IT

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question NO.	Questions	Max. Marks																				
Q.1	Attempt any Four of the following.																					
(i)	A continuous random variable x has the p.d.f. defined by $f(x) = A + Bx$, $0 \leq x \leq 1$. If the mean of the distribution is $1/3$, find A & B .	24																				
(ii)	For normal distribution 30% items are below 45 & 8% items are above 64. Find the mean & standard deviation of the normal distribution.																					
(iii)	The ratio of the probability of 3 successes in 5 independent trials to the probability of 2 successes in 5 independent trials is $\frac{1}{4}$, find p & q .																					
(iv)	For a normal variate X with mean 25 & standard deviation 10, find the area between (a) $X=25$, $X=35$, (b) $X=15$, $X=35$																					
(v)	The number of accidents in a year attributed to taxi drivers in a city follows Poisson distribution with mean 3. Out of 1000 taxi drivers, find approximately the number of drivers with (i) no accident in a year (ii) more than 3 accident in a year																					
(vi)	Find Mean and variance of Binomial distribution.																					
Q.2	Attempt any Four of the following.																					
(i)	The average marks scored by 32 boys are 72 with a S.D. of 8, while that for 36 girls is 70 with a S.D. of 6. Test at 1% level of significance whether the boys perform better than girls.	28																				
(ii)	A certain injection administered to 12 patients resulted in the following changes of blood pressure: 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4. Can it be concluded that the injection will be in general accompanied by an increase in blood pressure?																					
(iii)	Theory predicts that the proportion of beans in the four groups A, B, C, D should be 9: 3: 3:1. In an experiment among 1600 beans the numbers in the four groups were 882, 313, 287, & 118. Does the experimental result support the theory?																					
(iv)	Table below shows the performances of student in mathematics and physics. Using Chi-square, test the hypothesis that the performance in mathematics is independent of performance in physics. <table><tr><td></td><td colspan="3">Grades in Mathematics</td></tr><tr><td></td><td>High</td><td>Medium</td><td>Low</td></tr><tr><td rowspan="3">Grades in physics</td><td>High</td><td>56</td><td>71</td><td>12</td></tr><tr><td>Medium</td><td>47</td><td>163</td><td>38</td></tr><tr><td>Low</td><td>14</td><td>42</td><td>81</td></tr></table>			Grades in Mathematics				High	Medium	Low	Grades in physics	High	56	71	12	Medium	47	163	38	Low	14	42
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Grades in physics	High	56	71	12																		
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	(v)	A drug was administered to 5 persons and the systolic blood pressure before and after was measured. The result are given below (Take 1% level of significance)											
		Candidates	I	II	III	IV	V						
		B.P. before	140	130	132	150	140						
		B.P. after	132	126	133	144	133						
		Test whether the drug is effective in lowering the systolic blood pressure.											
Q.3		Attempt any Two of the following											
	(i)	Obtain the Spearman's rank correlation coefficient between X&Y.											
		X	32	55	49	60	43	37	43	49	10	20	
		Y	40	30	70	20	30	50	72	60	45	25	
	(ii)	The equations of two lines of regression are $3x + 2y = 26$ and $6x + y = 31$ find (a) Means of x and y (b) Coefficient of Correlation between x and y (c) σ_y if $\sigma_x = 3$										12	
	(iii)	Fit a straight line $y = ax + b$ to the following data											
		x	1	2	3	4	5	6					
		y	49	54	60	73	80	86					
Q.4		Attempt any Four of the following.											
	i	Using forward difference formula find y when $x = 0.5$ from the following data.											
		X	0	1	2	3							
		Y	-1	1	1	-2							
	ii	Evaluate $\int_0^1 \frac{dx}{1+x}$ by using (i) Trapezoidal rule (ii) Simpson's (1/3)rd rule. Take $h = 0.25$.											
	(iii)	Solve using Gauss –seidel Method. Up to third iteration. $x + 7y - 3z = 22$ $5x - 2y + 3z = 18$ $2x - y + 6z = 22$										28	
	iv	Using Bisection Method find Positive root of $xe^x = 1$ lying between 0 and 1 up to two decimal places.											
	v	Express into factorial polynomial the function. $x^4 - 8x^3 + 18x^2 - 10x$											
Q.5		Attempt any one of the following.											
	(i)	Solve the following L.P.P. by Simplex method maximize $z = 5x_1 + 4x_2$ Subject to $6x_1 + 4x_2 \leq 24$ $x_1 + 2x_2 \leq 6$ $-x_1 + x_2 \leq 1$ $x_2 \leq 2$ where $x_1 \geq 0, x_2 \geq 0$											
	(ii)	Find (i) All basic solutions (ii) All feasible basic solutions (iii) Optimal feasible basic solution for the following L.P.P. maximise $z = 2x_1 - 2x_2 + 4x_3 - 5x_4$ Subject to $x_1 + 4x_2 - 2x_3 + 8x_4 \leq 2$ $-x_1 + 2x_2 + 3x_3 + 4x_4 \leq 1$ where $x_1, x_2, x_3, x_4 \geq 0$										8	